



Gatuline[®] Spot-Light

Age Spot Corrector



People make our name



Age spots, symptoms and solution



Age spots, known commonly as dark spots, are **permanent hyperpigmented** marks that appear on UV exposed areas of the skin. They are called age spots because their appearance **increases over time**.

Dark spots represent the **first sign of aging** in Asian skin. They are also one of the **main aging concerns** named by European and American women after wrinkles and sagging.

Photo-aged melanocytes suffer from an excessive dose of **free radicals** and **inflammation**. This triggers overproduction of melanin, resulting in its uneven distribution in the epidermis.

Up until recently, it was believed that the **melanogenesis** was **regulated** exclusively in the epidermis, by keratinocytes.

New research has demonstrated a **significant role of fibroblasts** in hyperpigmentation, in particular when the dermal environment is photo-degraded.

To improve treatment of age spots, it now appears to be **essential to treat both the epidermal and dermal causes**, and protect the extracellular matrix from photo-aging.

A double targeted action for higher efficacy

Gatuline® Spot-Light is the **first ingredient** that tackles age spots, preventively and curatively, via a targeted action at both **epidermal** and **dermal levels**.

- In the epidermis, Gatuline® Spot-Light **decreases melanin synthesis** via a strong **anti-tyrosinase activity**. It also **enhances desquamation**, eliminating accumulated melanin from the upper skin layers.
- In the dermis, Gatuline® Spot-Light **protects the fibroblast environment** from premature degradation, inhibiting the production of matrix metalloproteinases (MMPs) and protecting against **glycation**.
- Throughout the skin, Gatuline® Spot-Light **inhibits inflammation** and provides **protection from oxidative degradation**.

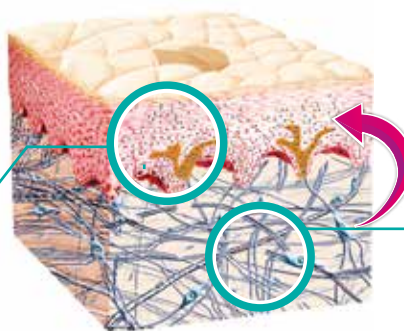
Throughout the skin

Anti-oxidation

Up to 75% DPPH radical scavenging activity (dose effect)

Anti-inflammatory effect

Inhibition of pro-inflammatory mediators release with 0.5% Gatuline® Spot-Light
Interleukine-6: 70%
Interleukine-8: 63%



Degradation of the dermal environment, by MMPs and glycation, triggers release of melanogenic soluble factors from fibroblasts that stimulate the melanocyte function.

In the epidermis



Exfoliation

Up to +81% proteolysis of corneodesmosin due to enhanced enzymatic activity of kallikrein-related peptidase 5

Anti-tyrosinase activity

More than 80% tyrosinase activity inhibition from 0.1% Gatuline® Spot-Light (dose effect)

Inhibition of melanogenesis

58% melanin synthesis inhibition with 0.5% Gatuline® Spot-Light (B16 melanocytes, dose effect)

In the dermis

Inhibition of MMPs enzymatic activities

MMP-1: 27%
MMP-2: 61%
MMP-3: 59%
MMP-9: 33%
MMP-12: 26%

Anti-glycation activity

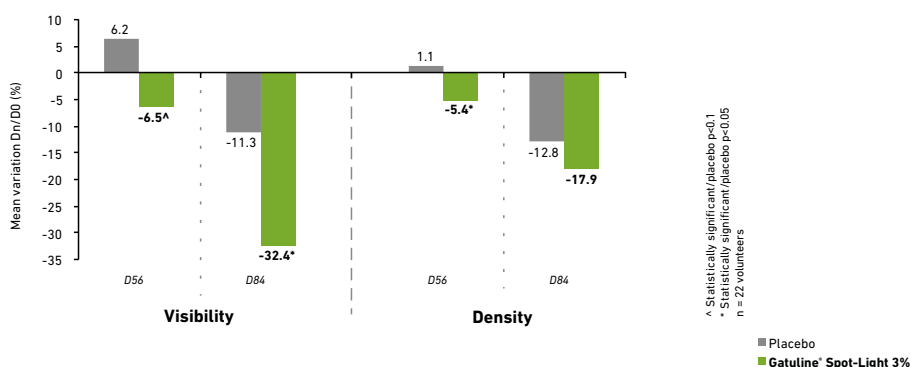
Up to 63% reduction of advanced glycation end products (dose effect)



Clinically proven

Gatuline® Spot-Light was tested *in vivo* with a **clinical study** ⁽¹⁾ versus placebo under dermatological control. The efficacy of the product was assessed by **clinical scoring**, **image analysis** of high definition photos, obtained under cross-polarized light, and a **use test questionnaire**.

Fewer, less visible age spots



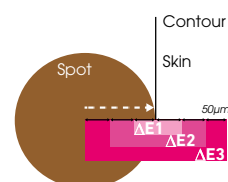
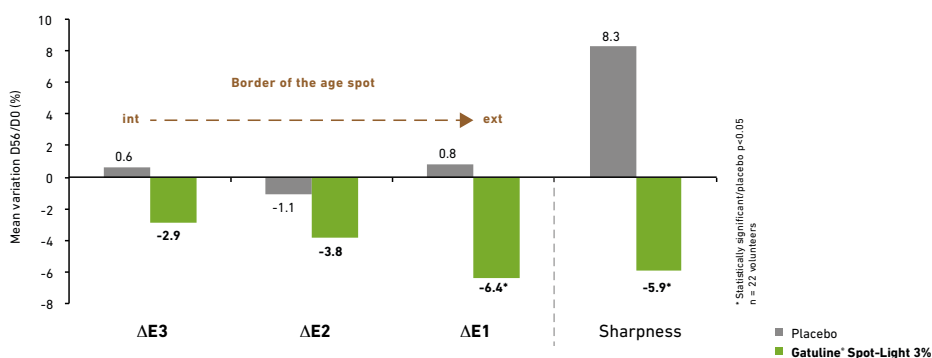
Visibility of the age spot: contrast between a spot and its immediate environment

Density of age spots: number of spots per unit area on the cheek

Clinical scoring ⁽²⁾

Gatuline® Spot-Light induces a significant reduction of the visibility and density of age spots.

Faded appearance



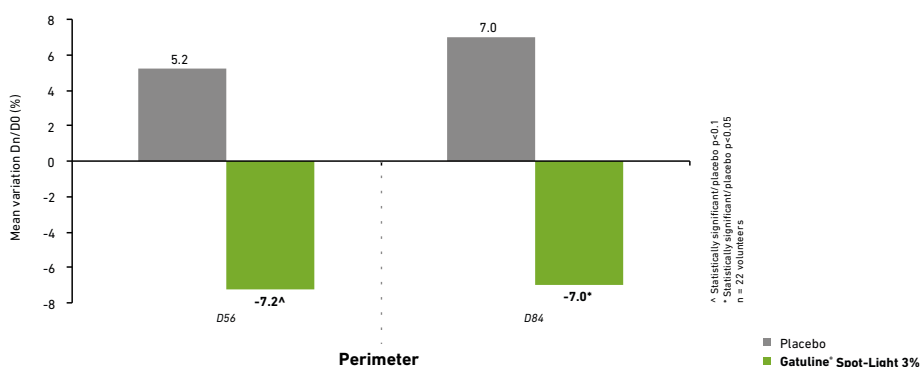
ΔE: color contrast between the age spot and normal adjacent skin ⁽³⁾

Sharpness: visibility of the age spot contour ⁽⁴⁾

Image analysis

Gatuline® Spot-Light blurs the contours of the age spot.

Smaller size



Perimeter: calculated in pixels

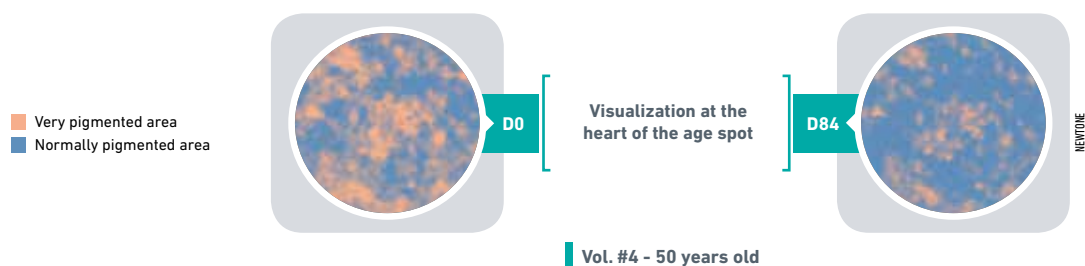
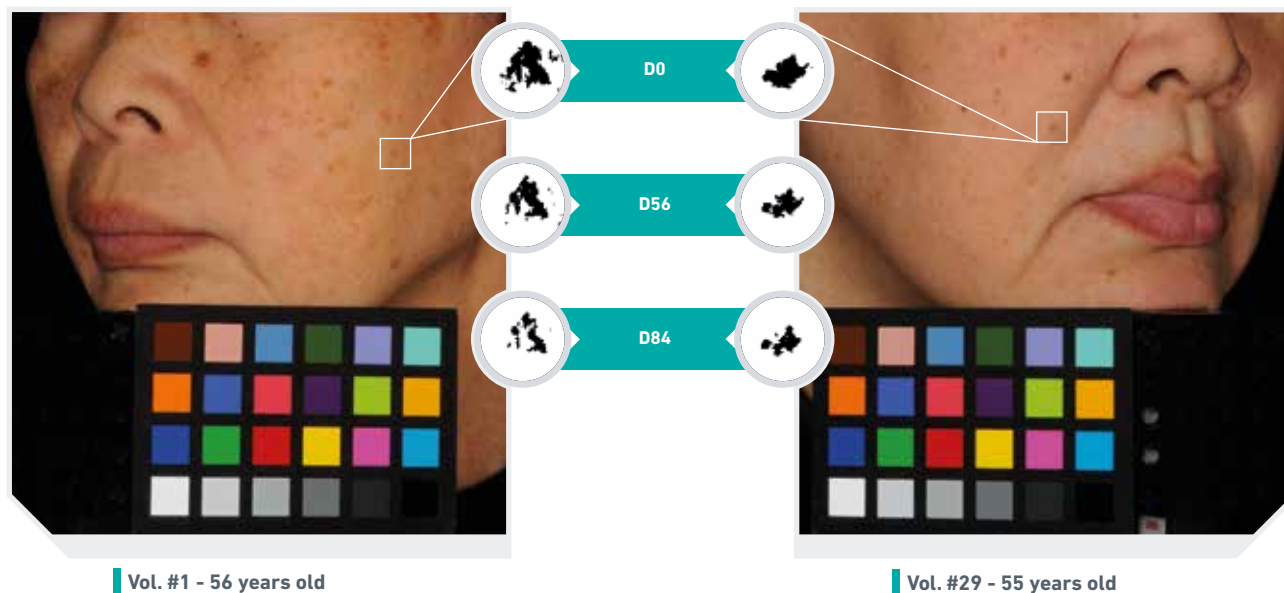


Image analysis

Gatuline® Spot-Light reduces significantly the perimeter of the age spot.

Visible efficacy

Gatuline® Spot-Light shows a **visible correcting action** on age spots of **different morphologies**.



High appraisal levels from users

The clinical efficacy of the product was recognized and **acclaimed by the panelists**. The skin looks **brighter, younger**, and age spots are **minimized**.



⁽¹⁾ 84-day clinical evaluation in Thailand, 30 volunteers, involving twice daily application to split face with untreated forehead of two creams (SPF 20) in a double blind study - One cream contained 3% Gatuline® Spot-Light, the other was a placebo cream - The panelists were aged from 45 to 63, all skin types, with light and intermediate skin color (mainly phototype III) and ITA° > 28° on the cheek
⁽²⁾ Clinical scoring, performed by a trained judge, based on Skin Aging Atlas, Asian Type, Vol. 2 (R. Bazin & F. Flamant)
⁽³⁾ $\Delta E = (\Delta L^* + \Delta a^* + \Delta b^*)^{1/2}$
⁽⁴⁾ Variance of ITA° (Individual Typological Angle)



Gatuline® Spot-Light



is a safe and natural solution for a brighter skin complexion with less visible age spots.

Gatuline® Spot-Light is obtained by extraction of active molecules from Asian *Sophora flavescens* and handpicked Italian kiwi fruit using natural, plant-derived solvents ⁽⁵⁾.

To ensure reproducible efficacy, the manufacturing process is tightly controlled and the content in flavanones and matrine, molecules holding strong biological activity, is guaranteed.

Sophora and Kiwi, are of well known composition and present an excellent safety record. Unlike kojic acid, Gatuline® Spot-Light is absolutely not cytotoxic at the same concentrations and presents good skin tolerance.

Claims

- Acts on both epidermal and dermal causes of age spots
- Reduces the visibility and number of age spots
- Fades the appearance of dark spots
- Lightens skin tone
- Safe and natural

Applications

- Age spot targeted treatments
- Skin perfectors, foundations and concealers
- Brightening products
- Alphabet creams
- Global anti-aging lines

Use level

3%

INCI name: Butylene Glycol (and) Actinidia Chinensis (Kiwi) Fruit Water (and) Alcohol (and) *Sophora Flavescens* Root Extract

Worldwide approval for cosmetic applications

Listed on IECIC (Inventory of Existing Cosmetic Ingredients in China) 2014-06

ECOCERT ⁽⁶⁾ and COSMOS ⁽⁷⁾ compliant, NPA™ certified

⁽⁵⁾ Vegetable origin butylene glycol

⁽⁶⁾ The product has been validated by ECOCERT Greenlife according to the Natural and Organic Cosmetics Standards

⁽⁷⁾ The product has been validated by ECOCERT Greenlife according to the COSMOS standard of Organic and Natural cosmetics



GATULINE® SPOT-LIGHT-GE-8A3-1112 (5893BE)

www.gattefosse.com



Corporate Headquarters

36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - France
+(33) 4 72 22 98 00

People make our name